VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

Ho Chi Minh City University of Technology

Faculty of Computer Science and Engineering



Deep Learning and Its Applications (CO3133)

Assignment 1 Report

Foundations of Deep Learning Pipelines and Architectures

From Linear Models to Modern Sequence Models:
A Comparative Study for Image Classification

Instructor: Le Thanh Sach

Semester: 261
Group: 18 – DeepDive
Students: Le Pham Tien Long – 2352688
Ngo Tieu Nghi – 2352799
Ho Minh Nhat – 2352858

HO CHI MINH CITY, 2026

Contents

List of Figures

List of Tables

1 Problem and Data Description

1.1 Problem formulation

1.2 Dataset description

1.3 Exploratory data analysis

1.4 Data splitting

1.5 Preprocessing

2 Methodology

2.1 Pipeline overview

2.2 Baseline and main methods

2.3 Training setup

2.4 Experimental design

3 Implementation Results

3.1 Training behaviour

3.2 Quantitative results

3.3 Qualitative results

3.4 Comparison and discussion

3.5 Error analysis

3.6 Ablation study

3.7 Limitations and conclusion

4 AI Usage Disclosure

Code Availability

The implementation, configurations and committed splits are available at `https://github.com/longlephamtien/DL261-Assignment`, and the assignment page at `https://longlephamtien.github.io/DL261-Assignment/assignments/assignment-1/`.